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ResumeLens

TEACHER

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1. Introduction, The problem

Recruitment processes often require reviewing large numbers of résumés and also the same qualification may appear using different names, abbreviations, or spelling conventions. In addition, different professional profiles require different combinations of qualifications.

That is why we are asked to develop ResumeLens an application that processes textual résumés and determines whether the qualifications identified in a candidate's résumé satisfy patterns defined for a professional profile.





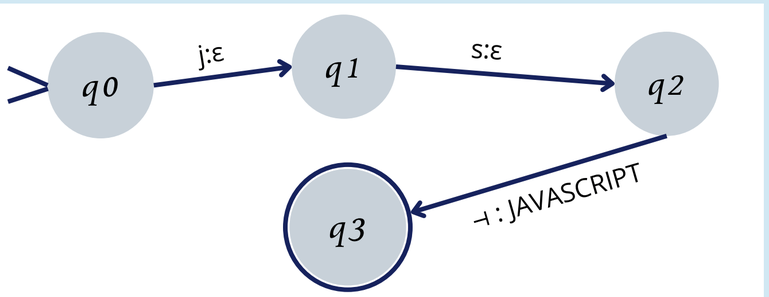
2. Metodology

First, we have to extract and separate the relevant information from the résumés. We did it by using regular expressions

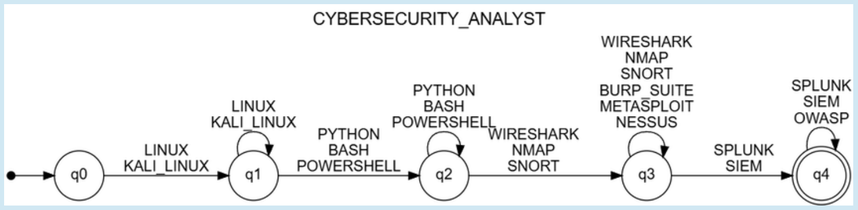
```
email_match = re.search(r'[\w.-]+@[\w.-]+\.\w+', text)
```

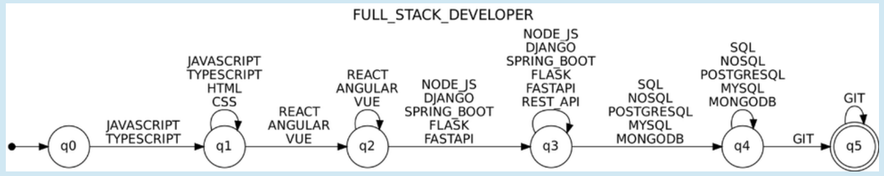
```
pattern = (
        r'((?:Bachelor|Master|PhD|Associate)(?: of [A-Za-z]+)?)'
        r' in ([A-Za-z\s]+?),\s*([A-Za-z0-9\s.&]+?),\s*(\d{4})'
    )
```

Part 2: Candidates express the same skills using abbreviations, typos, or synonyms. We implemented deterministic Finite-State Transducers :

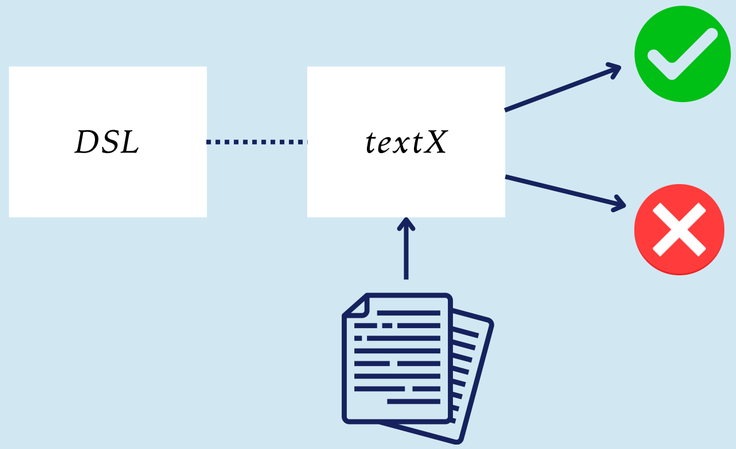


Part 3: Candidates may possess different combinations of skills depending on their professional profiles. We implemented Finite Automata to recognize qualification patterns and determine whether candidates satisfy the requirements defined for each profile.



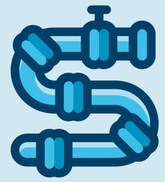


Part 4: Candidate information and classification results must follow a structured format. We implemented Context-Free Grammars using textX to validate candidate profiles and generate structured representations for visualization.



03. System Architecture

ResumeLens is organized into four processing stages: information extraction, qualification normalization, profile-pattern recognition, and candidate-profile validation. Each module transforms the output of the previous stage into a representation suitable for the next one.



04. Limitations

The system relies on predefined regular expressions, qualification aliases, profile patterns, and grammar rules. Consequently, unfamiliar terminology or résumé formats may not be recognized correctly. Future improvements could expand the qualification catalog, support additional input formats, and evaluate the system using a broader collection of résumés.



05. Conclusion

ResumeLens demonstrates how formal language models can be integrated into a modular résumé-processing pipeline. Regular expressions, finite-state transducers, finite automata, and context-free grammars address distinct processing tasks, from extracting textual information to validating structured candidate profiles.